HABIB UNIVERSITY

EE-172/CS-130/ CE-222 DIGITAL LOGIC AND DESIGN

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DIGITAL SYSTEM DESIGN CHALLENGE (SPRING-25)

OBJECTIVE: The aim of this project is to design and build a digital interaction device on an FPGA board.

What do we mean by interaction?

> "Interaction: a cyclic process in which two or more actors alternately listen, think, and speak."
>
> -Chris Crawford, *Designing Interactivity*

One of the actors in each of your project ideas will certainly be the system that you'll build and the other actor(s) will be humans. Note that an interaction requires communication between the actors, and so your system should be able to respond meaningfully to an interacting user for a period of time.

PROBLEM DESCRIPTION: DLD course project is your opportunity to design, build, debug, demonstrate, and report on a relatively small digital system. This document sets forth expectations and requirements for this project and makes a few suggestions which should help to make your project a success. In order to accomplish all that is expected by the end of the term, it is essential that you stay on schedule.

As an interaction device, your system should be able to alternately listen, think, and respond. So, your system is required to have three key components, corresponding to each of these tasks, as shown in Figure 01 and described below:

- **FPGA Board (BASYS 03)**: The brain (thinking) part of your system is to be implemented on a BASYS 03 FPGA board. You are ***required*** to write code using Verilog HDL, synthesize, and implement all the required logic and processing on BASYS 03 FPGA board yourself. The brain of your system is required to be modeled as a deterministic finite state machine (FSM).
  See Tutorial Videos on implementation of Verilog modules on FPGA boards.

- **Output Peripheral**: Your system is required to be able to respond to the interacting user in some way. This could be in the form of text/images on a display, LEDs, sound, 7-segment displays, motion, USB port etc. The BASYS 03 FPGA board has a number of in-built options such as LEDs, 7-segment displays, and VGA connector.

The lab 11 will provide you with basic understanding and relevant skills to use the VGA connector, the most complex of these output peripherals.

- **Input Peripheral(s):** In order for you to be able to listen to the user, you are required to have at least one input peripheral in your system. The BASYS 03 FPGA board provides you with in-built options like slide switches and push buttons, which can be expanded to other options such as sensors, mouse, joystick, etc. using USB port.
  Students are encouraged to explore interfacing of sensors or design custom consoles.

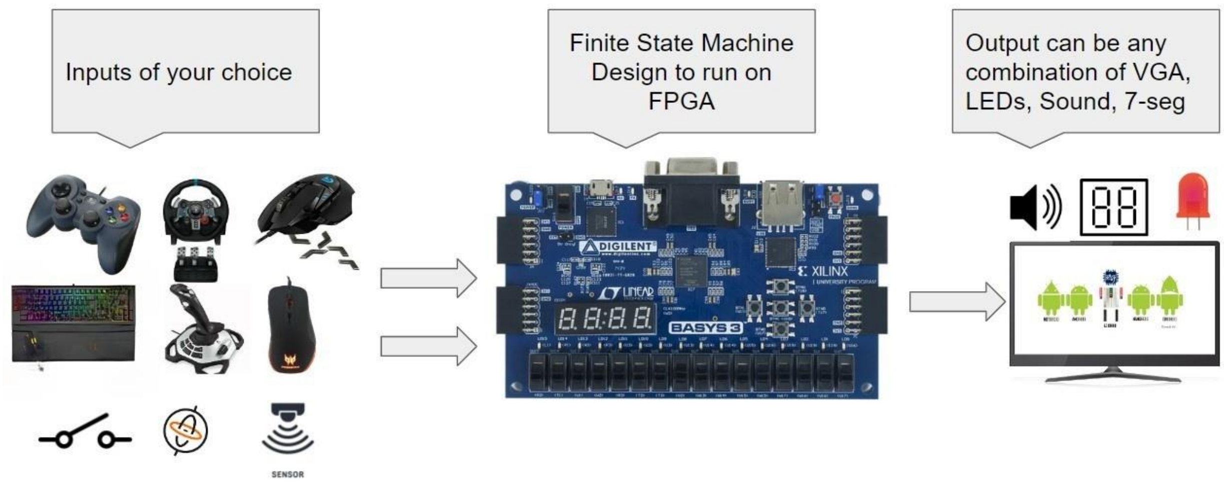

Figure 01: Overview of Project Design

Complexity Expectations:

Minimum requirement of project is to build at least three finite state machines i.e., one FSM to implement mode of operation, one FSM to implement VGA controller and one major FSM based on application chosen. You are encouraged to be creative in your project design and application. For application you can design any arcade game, a learning or educational assistant or any other interactive and interesting project which satisfies project requirements. Top three projects will be recognized and awarded in DPEC [DLD Project Exhibition and Competition] to be held at the end of semester.

Examples:

Note: These examples just show the capabilities of FPGA implementations. These may or may not meet project requirements of Fall-23.

1. *[Video]* Ping Pong
2. *[Video]* Two player Tank Game[Fall-20]
3. *[Video]* Snake Game [Fall-20]
4. *[Video]* Don't hit the bars
5. *[Video]* Radar
6. *[Video]* Game of Thrones theme
7. *[Image]* Piano Instructor [Fall-20]

8. *[Image]* Space Invaders [Fall-20]
9. *[Image]* DilDrop [Fall-20]

More examples will be shared in FAQs on LMS.

PROBLEM GOALS:

The purpose of this project is to help you obtain necessary skills to develop a hardware-software product-oriented project. In particular, you will acquire the following skillset after completing this project:

1. Programming digital logic on FPGA.
2. Design of logic modules involving comparators, multiplexers, encoders, decoders, counters, flip flops and finite state machines
3. Hardware/Software interfacing.
4. Hardware/Software debugging.
5. Interfacing of hardware components i.e., input/output peripherals with FPGA
6. Application specific logic development

INSTRUCTIONS:

1. Project work will be carried out in teams of up to minimum 04 students and maximum 05 students.
2. Teams will be formed within lab sections and must be approved by corresponding Lab RA/Instructor
3. Each team can only get one FPGA Development board issued to the team. It will be the responsibility of students to properly use the equipment and return the board at the end of Project demonstration in working condition.
4. For coding and design of Verilog modules, students can use EDA Playground or Xilinx Vivado Webpack.
5. For synthesis and testing, Xilinx Vivado will be needed. Therefore, it is recommended to install the required tool chain.
6. It is a digital hardware/software designing project for EE, CE and CS, and students from all three disciplines have to put equal efforts in it. However, it is recommended to have team members from different disciplines.

SCHEDULE:

Milestones	**Weightage** **T = 20%, L = 30%**	**Due on**
Milestone 0: Project kick-off: Team Formation	L = 1%	*Week 07*
Milestone 1: Project Proposal Submission	L= 4%	*Week 09*
Milestone 2: Early Prototype and Progress Report	L = 15%, T=10%	*Week 13*
Milestone 3: Final Prototype, Demo and Final Report	L = 10%, T=10%	TBA

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DLD Project Exhibition and Competition		TBA

T = Theory Grade, L = Lab Grade, TBA = To be announced

MILESTONE DESCRIPTION AND DELIVERABLES[1]:

Milestone 0: Team formation
Expected Deliverables: [L] Find team members, register your team and get it approved from lab RA.

Milestone 1: Project Proposal Submission[2]
Expected Deliverables: [L] Team submits proposal document which proposes two unique project ideas (A & B) which meet project requirements. Idea A would be a major project idea and idea B would be a backup idea in case idea A is found not to satisfy project requirements or it is beyond the scope of course.

Milestone 2: Early Prototype and Progress Report
Expected Deliverables:
1. [L] Demonstration of display of any graphic picture (say DLD Logo) on VGA display
2. [L] Demonstration of interfacing of selected input peripherals.
3. [T] Progress report on description/layout of project, type of FSM to be used, state definitions, state transition diagrams for system and results of (1) and (2)

Milestone 3: Final Prototype, Demo and Final Report
Expected Deliverables:
1. [L] Working demonstration and viva of project design
2. [T] Report describing working of the project with the help of block diagrams, logic diagrams, state diagrams etc. and listing of results and codes.
3. [T] 05 min video submission for DPEC

In case of queries/concerns contact your instructors/RAs. You can also post questions through Discussions on.

Good luck and have fun!

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[1] Checklist for milestones and relevant rubrics will be shared in week 08

[2] Template for project proposal will be shared in week 08